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METHOD FOR PRODUCING A VEHICLE BODY

Abstract

A method of manufacturing a vehicle body includes (i) casting an automotive support structure that defines a front body structure and a rear underbody structure; (ii) physically splitting the automotive support structure to detach the front body structure from the rear underbody structure; (iii) securing the front body structure to a rocker panel along a front end of the rocker panel after the splitting; and (iv) securing the rear underbody structure to the rocker panel along a rear end of the rocker panel after the splitting such that the front body structure is spaced-apart from the rear underbody structure via the rocker panel.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to vehicle body components and methods of making vehicle body components.

BACKGROUND

[0002] Vehicles and automobiles include bodies and/or frames that operate as the underlying support structure for the other subsystems of the vehicle (e.g., powertrain systems, steering systems, etc.).

SUMMARY

[0003] A method of manufacturing a vehicle body includes (i) casting an automotive support structure; (ii) physically splitting the automotive support structure to define a front body structure and a rear underbody structure; (iii) directly securing the front body structure to one or more additional vehicle body subcomponents; and (iv) directly securing the rear underbody structure to the one or more additional vehicle body subcomponents. The front body structure, the rear underbody structure, and the one or more additional vehicle body subcomponents form the vehicle body upon securing the rear underbody structure to the one or more additional vehicle body subcomponents. At least some of the one or more additional vehicle body subcomponents are positioned between the front body structure and the rear underbody structure upon securing the rear underbody structure to the one or more additional vehicle body subcomponents. The front body structure and the rear underbody structure are spaced-apart from each other within the formed vehicle body.

[0004] A method of manufacturing a vehicle body includes (i) casting an automotive support structure that defines a front body structure and a rear underbody structure; (ii) physically splitting the automotive support structure to detach the front body structure from the rear underbody structure; (iii) securing the front body structure to a rocker panel along a front end of the rocker panel after the splitting; and (iv) securing the rear underbody structure to the rocker panel along a rear end of the rocker panel after the splitting such that the front body structure is spaced-apart from the rear underbody structure via the rocker panel.

[0005] A method of manufacturing a vehicle body includes (i) casting a front body structure and a rear underbody structure as a first integrated structure; (ii) cutting the first integrated structure to separate the front body structure from the rear underbody structure; (iii) securing the front body structure to first regions of a pair of opposing rocker rails; and (iv) securing the rear underbody structure to second regions of the pair of opposing rocker rails such that the front body structure, the rear underbody structure, and the pair of opposing rocker rails form a second integrated structure. The front body structure and the rear underbody structure are spaced-apart from each other within the second integrated structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a side view of a central portion (e.g., a cabin portion) and a rear portion (e.g., a trunk portion) of a vehicle body structure;

[0007] FIG. **2** is a perspective view of a front body structure, which forms a front portion of the vehicle body structure;

[0008] FIG. **3** is a perspective view of a rear underbody structure, which partially forms a rear portion of the vehicle body structure;

[0009] FIG. **4** is a perspective view of a lower portion or lower structure of the vehicle body that includes the front body structure and the rear underbody structure;

[0010] FIG. **5** is a schematic diagram illustrating a process for forming the front body structure and the rear underbody structure; and

[0011] FIG. **6** is a flowchart illustrating a method for forming the vehicle body structure.

DETAILED DESCRIPTION

[0012] Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0013] Referring to FIG. **1** a side view of a portion of a vehicle body structure **10** is illustrated. The vehicle body structure **10** may simply be referred to as the vehicle body. The vehicle body structure **10** may include roof rails **12**, A-pillars **14**, B-pillars **16**, C-pillars **18**, D-pillars **20**, rocker panels **22**, A-pillar tower reinforcements **24**, cross members, floor members, floor panels, or any other component or subcomponent of a vehicle body structure. The components or subcomponents of the vehicle body structure **10** may form front door frames **26** and rear door frames **28**. The image in FIG. **1** is illustrative of common vehicle body components or subcomponents. The specific shape and structure of the vehicle body **10** is not intended to be limiting. Other vehicle body structure configurations should be construed as encompassed herein. For example, some vehicle body structures may not include a D-pillars **20** or may not include rear door frames **28**.

[0014] The subcomponents of the vehicle body structure **10** may be secured to each other via fasteners (e.g., screws, rivets, bolts, etc.), adhesives, welding, or any other method known in the art. The subcomponents of the vehicle body structure **10** may be made from steel, aluminum, magnesium, titanium, or any other material that may operate as a support structure for the other components and subsystems of a vehicle.

[0015] Referring to FIGS. **2** and **5** a perspective view of a front portion or a front body structure **30** of a vehicle body structure (e.g., vehicle body structure **10**) and a schematic diagram illustrating a process for forming the front body structure **30** along with a rear underbody structure **32** are illustrated, respectively. The front body structure **30** and the rear underbody structure **32** are illustrated as bottom views in FIG. **5**.

[0016] The front body structure **30** may include first and second opposing inner front rails **34**. The first and second opposing inner front rails **34** may also be referred to as a pair of front rails **34**. The inner front rails **34** partially form the front body structure **30**. The inner front rails **34** define an engine compartment **36** therebetween. More specifically the inner front rails **34** may define opposing sides of the engine compartment **36**. The engine compartment **36** is a space that is provided to receive an engine or other power source (e.g., electric motor) for a vehicle that includes the front body structure **30**. The front body structure **30** may also include a crossmember **38** extending between the inner front rails **34**. The crossmember **38** also partially forms the front body structure **30**. The crossmember **38** may partially define the engine compartment **36**. More specifically, the crossmember **38** may define a rear or back of the engine compartment **36**.

[0017] The front body structure **30** may also include first and second outer shotgun rails **40**. The first and second outer shotgun rails **40** may also be referred to as a pair of shotgun rails **40**. The outer shotgun rails **40** also partially form the front body structure **30**. The first outer shotgun **40** rail is disposed transversely outward from the first inner front rail **34** relative to the engine compartment **36** while the second outer shotgun rail **40** is disposed transversely outward from the second inner front rail **34** relative to the engine compartment **36**.

[0018] The front body structure **30** may also include first and second wheel well panels **42** that define wheel wells **44** configured to receive the wheels of the vehicle that includes the front body structure **30**. The first and second wheel well panels **42** may also be referred to as a pair of wheel well panels **42**. The first wheel well panel **42** is disposed between the first inner rail **34** and the first outer shotgun **40** rail. The second wheel well panel **42** is disposed between the second inner rail **34** and the second outer shotgun **40** rail. The first and second wheel well panels **42** define first and second openings **46** configured to received shock or strut towers **48** (e.g., FIG. **4**). The first opening **46** is disposed between the first inner rail **34** and the first outer shotgun **40** rail. The second opening **46** is disposed between the second inner rail **34** and the second outer shotgun **40** rail.
[0019] Referring to FIGS. **3** and **5** a perspective view of the rear underbody structure **32** of the vehicle body structure (e.g., vehicle body structure **10**) and a schematic diagram illustrating a process for forming the front body structure **30** along with a rear underbody structure **32** are illustrated, respectively.
[0020] The rear underbody structure **32** may include first and second opposing rear side rails or rear side members **50**. The first and second opposing rear side rails or rear side members **50** may also be referred to as a pair of rear side rails or rear side members **50**. The first and second opposing rear side members **50** partially form the rear underbody structure **32**. The first and second opposing rear side members **50** partially define first and second wheel wells **52**, respectively. The rear underbody structure **32** may also include at least one crossmember **54** extending between the first and second opposing rear side members **50**. The at least one crossmember **54** also partially forms the rear underbody structure **32**.
[0021] Referring to FIG. **4**, a lower portion **56** of a vehicle body structure (e.g., vehicle body structure **10**) is illustrated. The lower portion **56** of a vehicle body structure includes the front body structure **30** and the rear underbody structure **32** along with additional vehicle body subcomponents that form the vehicle body, or more specifically that form the lower portion **56** of the vehicle body. The additional vehicle body subcomponents may include first and second strut towers **48**, first and second rocker rails or rocker panels **58**, and a trunk support structure **60**. The first and second strut towers **48** may also be referred to as a pair of strut towers **48**. The first and second rocker rails or rocker panels **58** may also be referred to as a pair of rocker rails or rocker panels **58**. Each of the first and second strut towers **48** may be secured to the front body structure **30**. The first and second strut towers **48** may be disposed within first and second openings **46**, respectively.
[0022] The front body structure **30** may be secured to the additional vehicle body subcomponents, or more specifically to the first and second rocker panels **58**, along front ends **62** of the first and second rocker panels **58**. More specifically, rear outward ends of the first and second inner front rails **34** may be secured to the front ends **62** of the first and second rocker panels **58**. The rear underbody structure **32** may be secured to the additional vehicle body subcomponents, or more specifically to the first and second rocker panels **58**, along rear ends **64** of the first and second rocker panels **58**. More specifically, front ends of the first and second opposing rear side members **50** may be secured to the rear ends **64** of the first and second rocker panels **58**.
[0023] A cabin space **66** may be defined between the front body structure **30**, rear underbody structure **32**, and the first and second rocker panels **58**. A bulkhead **68** may extend between the rear ends **64** of the first and second rocker panels **58**. The bulkhead **68** may be disposed over a forward crossmember of the at least one crossmember **54**.
[0024] The trunk support structure **60** includes first and second rearward side rails **70**, a trunk floor support **72**, and a rear support member or bumper support **74**. The first and second rearward side rails **70** may also be referred to as a pair of rearward side rails **70**. Front ends of the first and second rearward side rails **70** may be secured to rear ends of the first and second opposing rear side members **50**, respectively. The bumper support **74** may extend between and may be secured to rear ends of the first and second rearward side rails **70**. The trunk floor support **72** may be disposed between the first and second rearward side rails **70**. The trunk floor support **72** may extend between

a rearward crossmember of the at least one crossmember **54** and the bumper support **74**. The trunk floor support **72** may be secured to each of the rearward crossmember of the at least one crossmember **54** and the bumper support **74**.

[0025] The subcomponents of the lower portion **56** of the vehicle body structure may be secured to each other via fasteners (e.g., screws, rivets, bolts, etc.), adhesives, welding, or any other method known in the art. The subcomponents of the lower portion **56** of the vehicle body structure may be made from steel, aluminum, magnesium, titanium, or any other material that may operate as a support structure for the other components and subsystems of a vehicle.

[0026] Referring to FIG. **5**, a schematic diagram of a process for forming the front body structure **30** and the rear underbody structure **32** is illustrated. The process includes first casting an automotive support structure **76**. The automotive support structure **76** is then physically split along cut lines **78**, which defines or forms the front body structure **30** and the rear underbody structure **32**. The front body structure **30** and the rear underbody structure **32** may also be present or defined after the casting process within the automotive support structure **76**. The automotive support structure **76** may be referred to as a first unified or integrated structure that includes or encompasses the front body structure **30** and the rear underbody structure **32**.

[0027] Referring to FIG. **6**, a flowchart of a method **100** for forming a vehicle body structure (e.g., vehicle body structure **10**), or more specifically, the lower portion **56** of a vehicle body structure is illustrated. The method **100** begins at block **102** where the automotive support structure **76** is cast. The casting process may include poring molten material into a die, followed by allowing the molten material to solidify to form the automotive support structure **76**. The cast automotive support structure **76** may define the front body structure **30** and the rear underbody structure **32** at block **102**. Stated in the alternative, the front body structure **30** and a rear underbody structure **32** may be cast as a first integrated structure at block **102**.

[0028] Casting the automotive support structure **76** may include (i) forming the first and second opposing inner rails **34**, (ii) forming the crossmember **38** extending between the first and second opposing inner rails **34**, (iii) forming the first and second opposing outer shotgun rails **40**, (iv) forming the first and second wheel well panels **42**, (v) defining the first and second openings **46**, (vi) forming the first and second opposing rear side members **50**, and (vii) forming the at least one crossmember **54** extending between the first and second opposing rear side members **50**.

[0029] The method **100** next moves on to block **104**, where the automotive support structure **76** is physically split to define or form the front body structure **30** and the rear underbody structure **32** (e.g., the automotive support structure **76** is cut along line along cut lines **78**). The front body structure **30** and the rear underbody structure **32** may be defined within the automotive support structure **76**. Therefore, physically splitting the automotive support structure **76** at block **104** may include splitting or cutting the automotive support structure **76** (or the first integrated structure that includes the front body structure **30** and the rear underbody structure **32**) to detach or separate the front body structure **30** and the rear underbody structure **32** from each other. The automotive support structure **76** may be physically split during a degating process where excess material is removed from the automotive support structure **76** after the casting process.

[0030] Next, the method **104** moves on to block **106**, where the front body structure **30** is directly secured to one or more additional vehicle body components (e.g., any of the components illustrated in FIG. **4**). More specifically, the front body structure **30** may be secured to first regions of the first and second rocker panels **58**. The first regions of the first and second rocker panels **58** may correspond to the front ends **62** of the first and second rocker panels **58**. The front body structure **30** may be secured to the front ends **62** of the first and second rocker panels **58** in the same manner as described and illustrated with respect to FIG. **4**.

[0031] The method next moves on to block **108**, where the rear underbody structure **32** is directly secured to one or more additional vehicle body components (e.g., any of the components illustrated in FIG. **4**). More specifically, the rear underbody structure **32** may be secured to second regions of the first and second rocker panels **58**. The second regions of the first and second rocker panels **58** may correspond to the rear ends **64** of the first and second rocker panels **58**. The rear underbody structure **32** may be secured to the rear ends **64** of the first and second rocker panels **58** in the same manner as described and illustrated with respect to FIG. **4**. It is noted that the chronological order of block **106** and block **108** may be changed so that block **108** occurs before block **106**. It is also noted that the steps in blocks **106** and **108** may occur simultaneously.

[0032] According to the steps in blocks **106** and **108**, the front body structure **30** and the rear underbody structure **32** are secured to the one or more additional vehicle body components (e.g., the first and second rocker panels **58**) such that (i) the front body structure **30**, the rear underbody structure **32**, and the one or more additional vehicle body subcomponents form the vehicle body (e.g., vehicle body structure **10**) or a portion of the vehicle body (e.g., lower portion **56** of a vehicle body structure), (ii) at least some of the one or more additional vehicle body subcomponents (e.g., the first and second rocker panels **58**) are positioned between the front body structure **30** and the rear underbody structure **32**, and (iii) the front body structure **30** and the rear underbody structure **32** are spaced-apart from each other within the formed vehicle body or the formed portion of the vehicle body. More specifically, the front body structure **30** and the rear underbody structure **32** may be spaced-apart from each other within the formed vehicle body or the formed portion of the vehicle body via the first and second rocker panels **58**. The front body structure **30**, the rear underbody structure **32**, and the one or more additional vehicle body components may be referred to as a second unified or integrated structure once the front body structure **30**, the rear underbody structure **32**, and the one or more additional vehicle body components are secured to each other at blocks **106** and **108**.

[0033] Although the steps illustrated in the blocks of FIG. **6** may generally be in chronological order, it should be understood that the flowchart in FIG. **6** is for illustrative purposes only and that the method **100** should not be construed as limited to the flowchart in FIG. **6**. Some of the steps of the method **100** may be rearranged while others may be omitted entirely.

[0034] Current production includes the utilization of a front structure casting and a rear underbody casting. These castings have replaced assemblies typically comprising many individual stampings, extrusions, or small castings that are then joined in a multi-stage process. The consolidation of these parts into a single cast component eliminates the assembly process and associated expense and capital footprint for assembly tooling. Although this consolidation is an overall savings, each die casting machine may require a large investment as multiple die casting machines may be required to support production volumes.

[0035] The system disclosed herein covers a proposed manufacturing process where the front structure casting and the rear underbody casting are cast together in a single casting press. During the degating process, where the runner system is typically removed from the parts, the front and rear underbodies are separated. Combining the front and rear underbody castings into a single tool drives slightly larger press sizes relative to casting front and rear underbody castings separately. However, the investment required for a larger press is more than offset by the expense reduction that results from cutting the total number of dies in half.

[0036] It should be understood that the designations of first, second, third, fourth, etc. for any component, state, or condition described herein may be rearranged in the claims so that they are in chronological order with respect to the claims. Furthermore, it should be understood that any component, state, or condition described herein that does not have a numerical designation may be given a designation of first, second, third, fourth, etc. in the claims if one or more of the specific component, state, or condition are claimed.

[0037] The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments may be combined to form further embodiments that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

Claims

1. A method of manufacturing a vehicle body comprising: casting an automotive support structure; physically splitting the automotive support structure to define a front body structure and a rear underbody structure; directly securing the front body structure to one or more additional vehicle body subcomponents; and directly securing the rear underbody structure to the one or more additional vehicle body subcomponents such (i) that the front body structure, the rear underbody structure, and the one or more additional vehicle body subcomponents form the vehicle body, (ii) at least some of the one or more additional vehicle body subcomponents are positioned between the front body structure and the rear underbody structure, and (iii) the front body structure and the rear underbody structure are spaced-apart from each other within the formed vehicle body.
2. The method claim 1, wherein physically splitting the automotive support structure to define the front body structure and the rear underbody structure comprises cutting the automotive support structure.
3. The method claim 1, wherein the one or more additional vehicle body subcomponents includes first and second rocker panels.
4. The method claim 3, wherein (i) the front body structure is secured to the one or more additional vehicle body subcomponents along front ends of the first and second rocker panels and (ii) the rear underbody structure is secured to the one or more additional vehicle body subcomponents along rear ends of the first and second rocker panels.
5. The method claim 1, wherein (i) casting the automotive support structure includes forming first and second opposing inner rails that partially form the front body structure and (ii) the first and second opposing inner rails define an engine compartment therebetween.
6. The method claim 5, wherein casting the automotive support structure includes forming a crossmember extending between the first and second opposing inner rails.
7. The method claim 5, wherein (i) casting the automotive support structure includes forming first and second opposing outer shotgun rails that partially form the front body structure, (ii) the first outer shotgun rail is disposed transversely outward from the first inner rail relative to the engine compartment, and the (iii) the second outer shotgun rail is disposed transversely outward from the second inner rail relative to the engine compartment.
8. The method claim 6, wherein (i) casting the automotive support structure includes defining first and second openings configured to received strut towers along the front body structure, (ii) the first opening is disposed between the first outer shotgun rail and the first inner rail, and (iii) the second opening is disposed between the second outer shotgun rail and the second inner rail.
9. The method claim 1, wherein (i) casting the automotive support structure includes forming first and second opposing rear side members that partially form the rear underbody structure and (ii) the first and second opposing rear side members partially define first and second wheel wells.
10. The method claim 9, wherein casting the automotive support structure includes forming at least one crossmember extending between the first and second opposing rear side members.
11. A method of manufacturing a vehicle body comprising: casting an automotive support structure that defines a front body structure and a rear underbody structure; physically splitting the automotive support structure to detach the front body structure from the rear underbody structure;

securing the front body structure to a rocker panel along a front end of the rocker panel after the splitting; and securing the rear underbody structure to the rocker panel along a rear end of the rocker panel after the splitting such that the front body structure is spaced-apart from the rear underbody structure via the rocker panel.

12. The method claim 11, wherein physically splitting the automotive support structure to detach the front body structure from the rear underbody structure comprises cutting the automotive support structure.

13. The method claim 11, wherein (i) casting the automotive support structure includes forming first and second opposing inner rails that partially form the front body structure and (ii) the first and second opposing inner rails define an engine compartment therebetween.

14. The method claim 13, wherein casting the automotive support structure includes forming a crossmember extending between the first and second opposing inner rails.

15. The method claim 13, wherein (i) casting the automotive support structure includes forming first and second opposing outer shotgun rails that partially form the front body structure, (ii) the first outer shotgun rail is disposed transversely outward from the first inner rail relative to the engine compartment, and the (iii) the second outer shotgun rail is disposed transversely outward from the second inner rail relative to the engine compartment.

16. The method claim 15, wherein (i) casting the automotive support structure includes defining first and second openings configured to received strut towers along the front body structure, (ii) the first opening is disposed between the first outer shotgun rail and the first inner rail, and (iii) the second opening is disposed between the second outer shotgun rail and the second inner rail.

17. The method claim 11, wherein (i) casting the automotive support structure includes forming first and second opposing rear side members that partially form the rear underbody structure.

18. The method claim 17, wherein the first and second opposing rear side members partially define first and second wheel wells.

19. The method claim 17, wherein casting the automotive support structure includes forming at least one crossmember extending between the first and second opposing rear side members.

20. A method of manufacturing a vehicle body comprising: casting a front body structure and a rear underbody structure as a first integrated structure; cutting the first integrated structure to separate the front body structure from the rear underbody structure; securing the front body structure to first regions of a pair of opposing rocker rails; and securing the rear underbody structure to second regions of the pair of opposing rocker rails such that the front body structure, the rear underbody structure, and the pair of opposing rocker rails form a second integrated structure, wherein the front body structure and the rear underbody structure are spaced-apart from each other within the second integrated structure.